Communication Systems Lecture-7: Digital Signaling

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Digital Signaling (1)

- Suppose that the message to be transmitted over a communication channel consists of N pieces of information (digital data) $\{w_k\}_{k=1}^{N}$.
- The analog signal at the output of the transmitter can be expressed as an orthogonal series with a finite number of terms (N):

$$w(t) = \sum_{k=1}^{N} w_k \phi_k(t) \quad ; \quad 0 \leq t \leq T_0$$

  - N is the number of dimensions needed to represent the signal over the interval $[0 \quad T_0]$.
  - $\{\phi_k(t)\}_{k=1}^{N}$ are N orthogonal functions.
  - $\{w_k\}_{k=1}^{N}$ constitute the digital data to be transmitted. w_k can take L possible values.
    - Example: with binary data, $L = 2$ and $w_k \in \{0, 1\}$.

Digital Signaling (2)

- For example, with the ASCII code, the letter X is assigned the codeword 0001101. In this case, $N = 7$ and:

$$w_1 = 0\,;\; w_2 = 0\,;\; w_3 = 0\,;\; w_4 = 1\,;\; w_5 = 1\,;\; w_6 = 0\,;\; w_7 = 1$$

- **Definition**: The baud rate (symbol rate) is:

$$D = \frac{N}{T_0} \quad (\text{symbols/s or bauds})$$

where N is the number of dimensions used in T_0 seconds.

- **Definition**: The bit rate is:

$$R = \frac{n}{T_0} \quad (\text{bits/s})$$

where n is the number of data bits sent in T_0 seconds.

- When the w_k's have binary values, $N = n$ and $w(t)$ is called a binary signal.
- When the w_k's are assigned more than two possible values, $w(t)$ is called a multilevel signal.

Digital Signaling (3)

At the receiver side, the w_k's need to be extracted from $w(t)$.

- Since $w(t)$ is an orthogonal series, data can be detected at the receiver by evaluating the orthogonal series coefficients:

$$w_k = \frac{1}{K_k} \int_0^{T_0} w(t)\phi_k^*(t)\mathrm{d}t \quad ; \quad k = 1 \cdots N$$

  In this case:

  - $w(t)$ is the waveform at the receiver.
  - $\{\phi_k(t)\}_{k=1}^N$ are the known orthogonal functions used to generate the transmitted waveform.

- It can be shown that the last equation is the optimal way to detect data when the received signal is corrupted by white additive noise. This procedure is called the matched filter detection.

Bandwidth Estimation

- From the dimensionality theorem, it can be proven that the bandwidth of $w(t)$ can be lower-bounded by:

$$B \geq \frac{N}{2T_0} = \frac{1}{2}D \ \ \text{Hz}$$

where D is the baud rate.

  - The lower bound of $\frac{1}{2}D$ is achieved if $\phi_k(t)$ are sinc-type functions.
  - Otherwise (for rectangular pulses for example), the bandwidth will be greater than this lower bound.
  - The above equation is useful to **approximate** the bandwidth of digital signals when the exact bandwidth is difficult to calculate.

Binary Signaling (1)

- Example: consider the case of binary signaling where 8 symbols are transmitted in 8 ms. Consequently:

$$N = 8 \quad ; \quad T_0 = 8 \text{ ms}$$

- In this case, each symbol corresponds to one bit:

$$n = N = 8$$

  - For example, if the codeword to be transmitted is 01001110:

$$w_1 = 0 \;\; ; \;\; w_2 = 1 \;\; ; \;\; w_3 = 0 \;\; ; \;\; w_4 = 0$$
$$w_5 = 1 \;\; ; \;\; w_6 = 1 \;\; ; \;\; w_7 = 1 \;\; ; \;\; w_8 = 0$$

- The symbol duration T_s which corresponds to the time needed to transmit 1 symbol is given by:

$$T_s = \frac{T_0}{N} = \frac{8 \text{ (ms)}}{8} = 1 \text{ ms}$$

Binary Signaling (2)

- Since the transmitted signal is binary, then the bit duration T_b which corresponds to the time needed to transmit 1 bit is given by:

$$T_b = T_s = 1 \quad \text{ms}$$

- Consequently, the baud rate is given by:

$$D = \frac{N}{T_0} = \frac{8\ (\text{symbols})}{8\ (\text{ms})} = 1 \quad \text{kbauds}$$

$$\text{or} \quad D = \frac{1}{T_s} = \frac{1\ (\text{symbol})}{1\ (\text{ms})} = 1 \quad \text{kbauds}$$

- In the same way, the data rate is given by:

$$R = \frac{n}{T_0} = \frac{8\ (\text{bits})}{8\ (\text{ms})} = 1 \quad \text{kbits/s}$$

$$\text{or} \quad R = \frac{1}{T_b} = \frac{1\ (\text{bit})}{1\ (\text{ms})} = 1 \quad \text{kbits/s}$$

Binary Signaling (3)

- Note that for binary signaling:

$$
\begin{aligned}
N &= n \\
T_s &= T_b \\
D &= R
\end{aligned}
$$

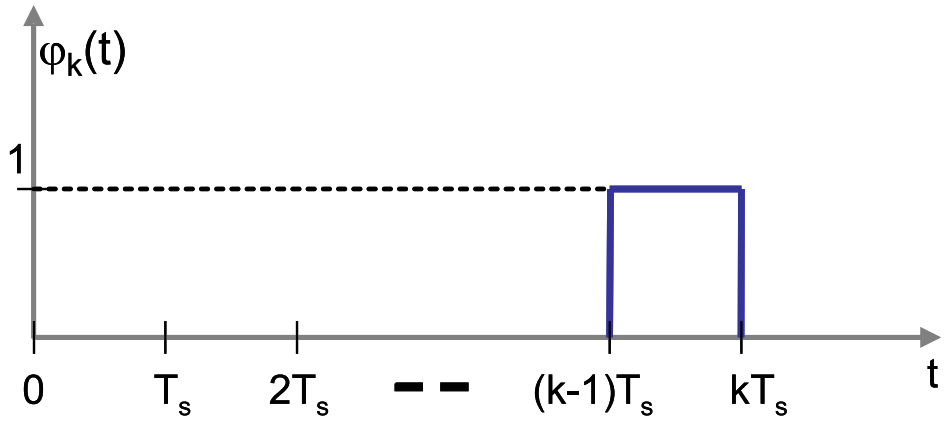
Binary Signaling (4)

- When rectangular-type functions are used, the waveform $w(t)$ is given by:

$$w(t) = \sum_{k=1}^{N=8} w_k \phi_k(t) \mathrm{d}t$$

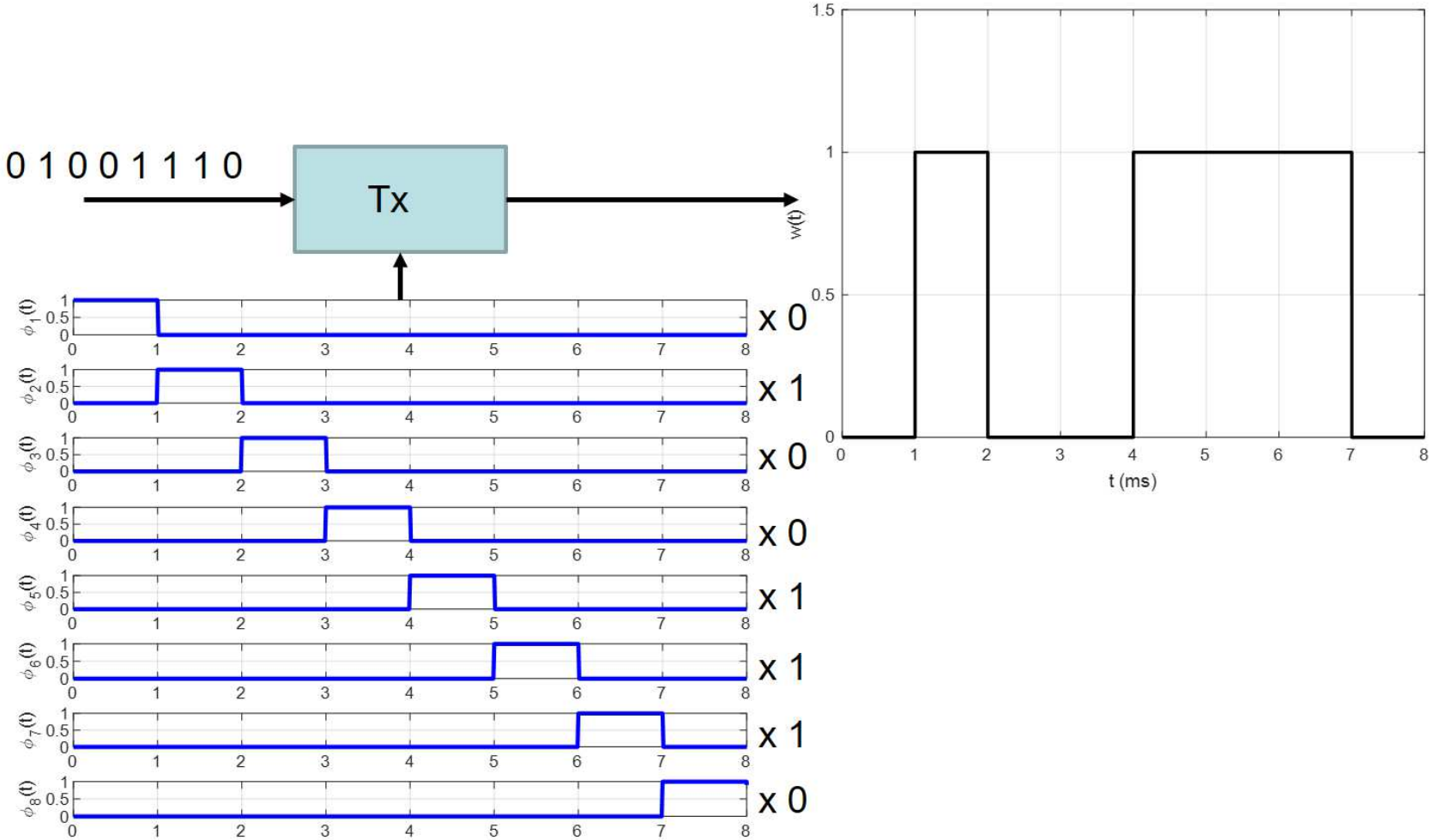
where:

$$\phi_k(t) = \mathrm{rect}\left[\frac{1}{T_s}\left[t - \left(k - \frac{1}{2}\right) T_s\right]\right] \quad ; \quad k = 1 \cdots N = 8$$



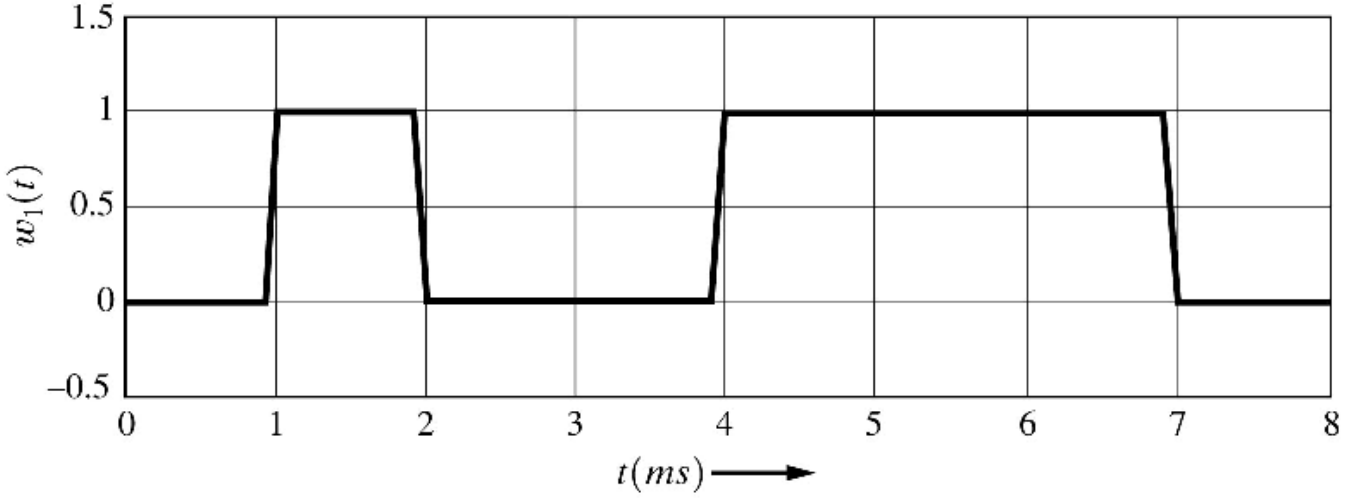
Binary Signaling (5)

- Consider the bit sequence 01001110. The implementation of $w(t) = \sum_{n=1}^{N} a_n \phi_n(t)$ is shown below (assuming $T_s = 1$):



Binary Signaling (6)

- In this case, the waveform corresponding to the bit sequence 01001110 is shown below:



(a) Rectangular Pulse Shape, $T_b = 1\ ms$

Binary Signaling (7)

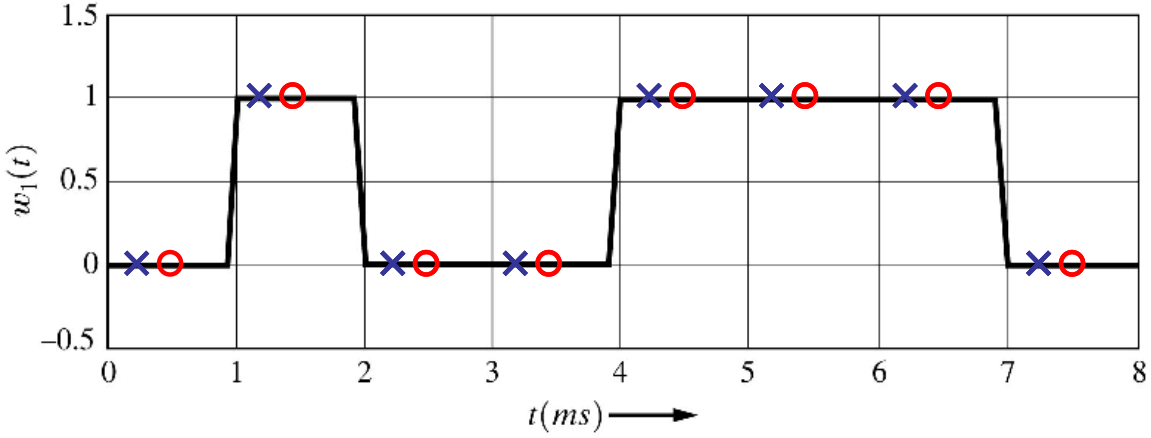
- The data can be detected at the receiver by evaluating the orthogonal-series coefficients:

$$w_k = \frac{1}{K_k}\int_0^{T_0} w(t)\phi_k^*(t)\mathrm{d}t \quad ; \quad k = 1 \cdots N = 8$$

  - where the constant K_k is given by:

$$K_k = \int_{-\infty}^{+\infty} |\phi_k(t)|^2 \mathrm{d}t = \int_{(k-1)T_s}^{kT_s} (1)^2 \mathrm{d}t = T_s$$

- For the case of rectangular pulse shapes, this is equivalent to sampling the waveform anywhere within each symbol interval:



(a) Rectangular Pulse Shape, $T_s = 1\ ms$

Binary Signaling (8)

- The Fourier transform of $\mathrm{rect}\left(\frac{t}{T_s}\right)$ is $T_s \mathrm{sinc}\left(T_s f\right)$.
- On the other hand, $\mathrm{sinc}\left(T_s f\right) = 0$ when f is a non-zero integer multiple of $\frac{1}{T_s}$.
- Consequently, the null-to-null bandwidth of the rectangular pulse shape is:

$$B = \frac{1}{T_s} = \frac{1}{1\ (\mathrm{ms})} = 1\ \mathrm{kHz}$$

- Note that this bandwidth is greater than the lower bound given by:

$$B_{L.B.} = \frac{1}{2} D = 500\ \mathrm{Hz}$$

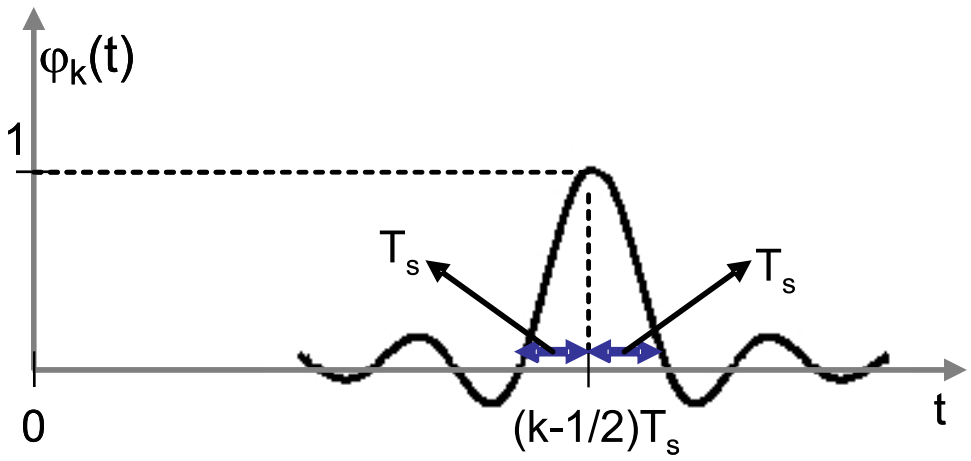
Binary Signaling (9)

- When sinc-type functions are used, the waveform $w(t)$ is given by:

$$w(t) = \sum_{k=1}^{N=8} w_k \phi_k(t) \mathrm{d}t$$

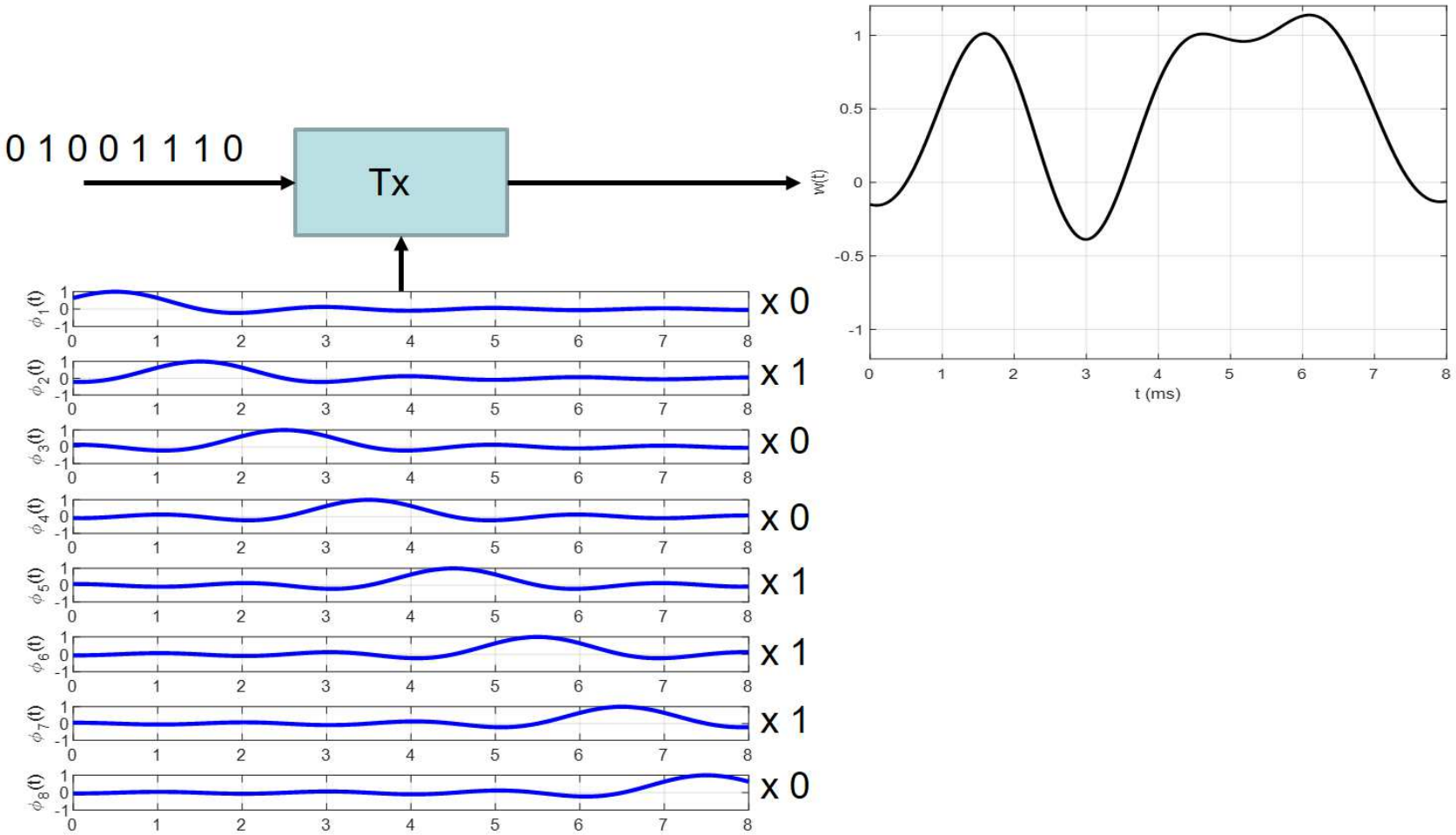
where:

$$\phi_k(t) = \mathrm{sinc}\left[\frac{1}{T_s}\left(t - \left(k - \frac{1}{2}\right) T_s\right)\right] \quad ; \quad k = 1 \cdots 8$$



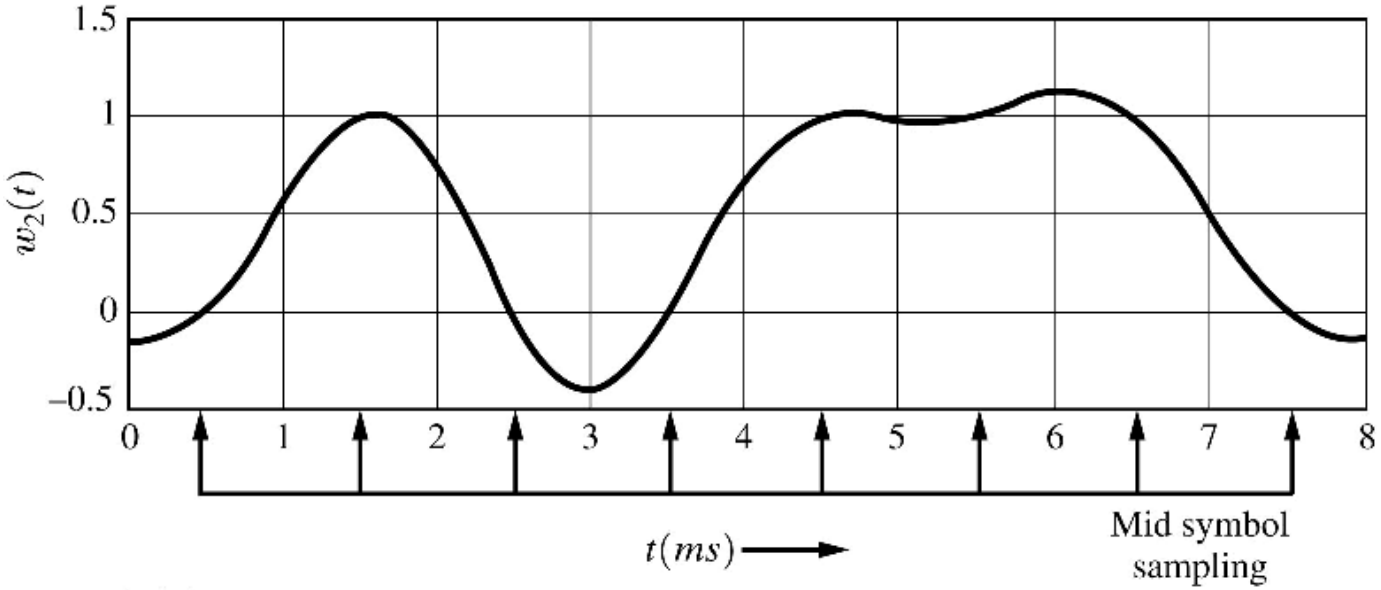
Binary Signaling (10)

- Consider the bit sequence 01001110. The implementation of $w(t) = \sum_{n=1}^{N} a_n \phi_n(t)$ is shown below (assuming $T_s = 1$):



Binary Signaling (11)

- In this case, the waveform corresponding to the bit sequence 01001110 is shown below:



(b) $\sin(x)/x$ Pulse Shape, $T_b = 1\ ms$

Binary Signaling (12)

- The data can be detected at the receiver by evaluating the orthogonal-series coefficients:

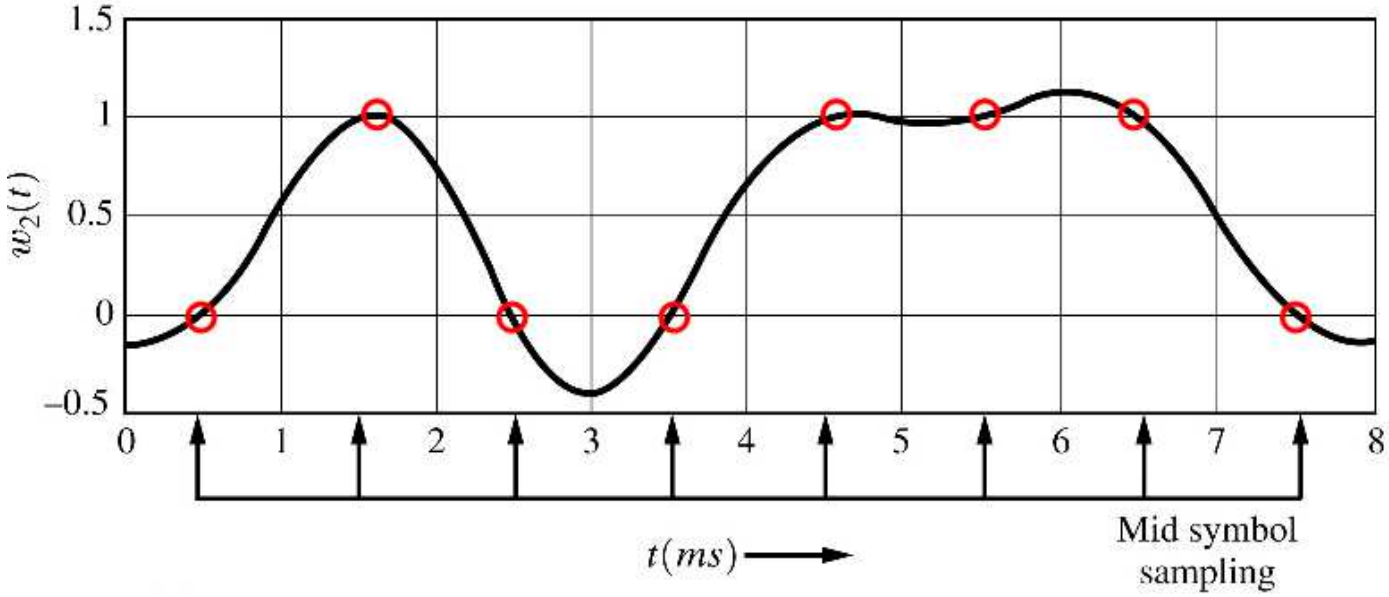
$$w_k = \frac{1}{K_k} \int_0^{T_0} w(t)\phi_k^*(t)\mathrm{d}t \quad ; \quad k = 1 \cdots N = 8$$

  - where the constant K_k is given by:

$$\begin{aligned} K_k &= \int_{-\infty}^{+\infty} |\phi_k(t)|^2 \mathrm{d}t = \int_{-\infty}^{+\infty} |\Phi_k(f)|^2 \mathrm{d}f \\ &= \int_{-\infty}^{+\infty} \left| T_s \mathrm{rect}(T_s f) e^{-j2\pi\left(k-\frac{1}{2}\right)T_s f} \right|^2 \mathrm{d}f \\ &= \int_{-\infty}^{+\infty} \left[T_s \mathrm{rect}(T_s f)\right]^2 \mathrm{d}f \\ &= T_s^2 \int_{-\frac{1}{2T_s}}^{+\frac{1}{2T_s}} (1)^2 \mathrm{d}f = T_s \end{aligned}$$

Binary Signaling (13)

- For the case of sinc-type pulse shapes, the data can be recovered by simply sampling the received waveform at the middle of each symbol duration.



Binary Signaling (14)

- The Fourier transform of $\mathrm{sinc}\left(\frac{t}{T_s}\right)$ is $T_s \mathrm{rect}\left(T_s f\right)$.
- On the other hand, $\mathrm{rect}\left(T_s f\right) = 0$ for $|f| > \frac{1}{2}\frac{1}{T_s}$.
- Consequently, the absolute bandwidth of the sinc-type pulse shape is:

$$B = \frac{1}{2T_s} = \frac{1}{2\ (\mathrm{ms})} = 500\ \mathrm{Hz}$$

- This bandwidth is equal to the lower bound given by:

$$B_{L.B.} = \frac{1}{2}D = 500\ \mathrm{Hz}$$

- Thus, the above lower bound can be achieved by sinc-type pulse shapes.

Multilevel Signaling (1)

Consider the case of a data source transmitting one of $M = 256$ possible messages.

- Suppose now that each symbol w_k can take $L > 2$ possible values.
- In other words, $l = \log_2(L)$ bits are grouped together in one symbol w_k.
- Consider the case where $L = 4$. The Gray mapping between the sequences of l bits and the L possible output levels is given by:

Binary Input (l=2 bits)	Output level (V)
11	+3
10	+1
00	-1
01	-3

Multilevel Signaling (2)

- In this case, the binary sequence of length $n = 8$ bits and whose value is 01001110 can be represented by $N = \frac{n}{l}$=4 multi-level symbols given by:

$$w_1 = -3 \;\; ; \;\; w_2 = -1 \;\; ; \;\; w_3 = +3 \;\; ; \;\; w_4 = +1$$

- In other words:

$$w(t) = \sum_{k=1}^{N} w_k \phi_k(t)$$

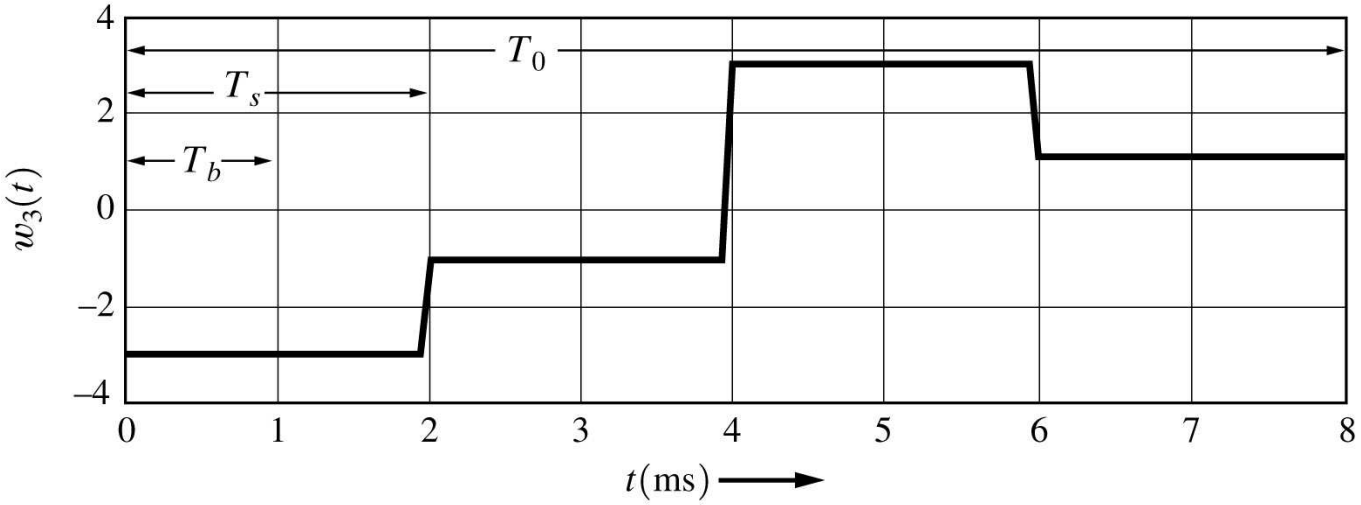
  - $\phi_k(t)$ can be a rectangular-type pulse shape:

$$\phi_k(t) = \mathrm{rect}\left[\frac{1}{T_s}\left[t - \left(k - \frac{1}{2}\right) T_s\right]\right] \;\; ; \;\; k = 1 \cdots N = 4$$

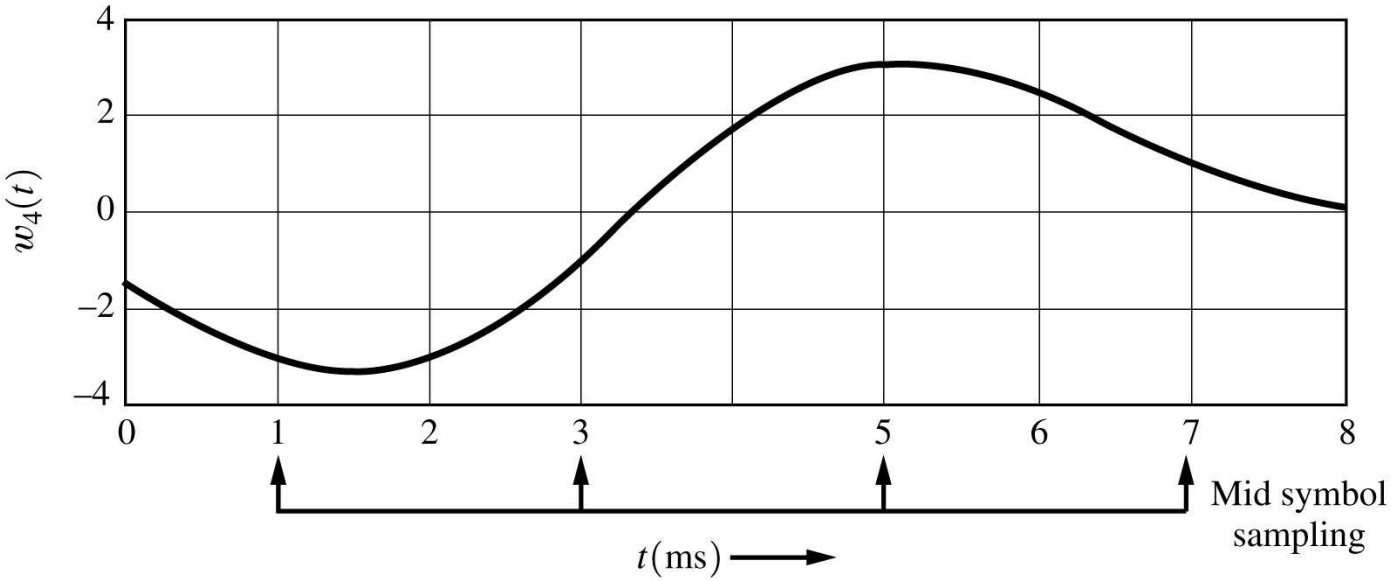
  - $\phi_k(t)$ can be a sinc-type pulse shape:

$$\phi_k(t) = \mathrm{sinc}\left[\frac{1}{T_s}\left(t - \left(k - \frac{1}{2}\right) T_s\right)\right] \;\; ; \;\; k = 1 \cdots N = 4$$

Multilevel Signaling (3)



(a) Rectangular Pulse Shape $T_b = 1$ ms



(b) $\sin(x)/x$ Pulse Shape, $T_b = 1$ ms

Multilevel Signaling (4)

- The symbol duration is given by:

$$T_s = \frac{T_0}{N} = \frac{8\ (\mathrm{ms})}{4} = 2\ \mathrm{ms}$$

- The bit duration is given by:

$$T_b = \frac{T_0}{n} = \frac{8\ (\mathrm{ms})}{8} = 1\ \mathrm{ms}$$

- Note that T_s and T_b are related to each other by the relation:

$$T_s = \frac{T_0}{N} = \frac{T_0}{n/\log_2(L)} = \log_2(L)\frac{T_0}{n} = \log_2(L)\,T_b$$

- This implies that the bit rate and baud rate are related to each other by:

$$R = \frac{1}{T_b} = \frac{1}{T_s/\log_2(L)} = \log_2(L)\frac{1}{T_s} = \log_2(L)D$$

  - In the previous example: $R = 1$ kbits/s while $D = 0.5$ kbaud.

Multilevel Signaling (5)

- The null-to-null bandwidth of the rectangular-type pulses is:

$$B = \frac{1}{T_s} = \frac{1}{2\ (\mathrm{ms})} = 500\ \mathrm{Hz}$$

- The absolute bandwidth of the sinc-type pulses is:

$$B = \frac{1}{2T_s} = \frac{1}{4\ (\mathrm{ms})} = 250\ \mathrm{Hz}$$

- As with binary signaling, the sinc-type functions achieve the lower bound of $B_{L.B.} = \frac{1}{2}D = 250$ Hz with multilevel signaling.

Multilevel Signaling (6)

- In general, the bandwidth with multi-level signaling is proportional to:

$$B_{\text{Multilevel}} = \frac{1}{T_s} = \frac{1}{\log_2(L)\,T_b} = \frac{1}{\log_2(L)} B_{\text{Binary}}$$

- The multilevel signal has $\frac{1}{\log_2(L)}$ the bandwidth of the corresponding binary signal.
- This bandwidth reduction is achieved because the symbol duration of the multilevel signal is $l = \log_2(L)$ times that of the binary signal.

Multilevel Signaling (7)

As with binary signaling:

- Rectangular pulse shapes: demodulation can be achieved by sampling the received waveform anywhere within each symbol interval (having a duration of T_s).
- Sinc-type pulse shapes: demodulation can be achieved by sampling the received waveform at the middle of each symbol interval (having a duration of T_s).